

51-mavzu. Ko'p o'zgaruvchili funktsiyaning yuqori tartibli hosila va differentsiallari. Teylor formulasi.

51-ma'ruza

Reja

- 1⁰. Yuqori tartibli xususiy hosilalar.
- 2⁰. Yuqori tartibli differentsiallar.
- 3⁰. Murakkab funktsiyaning yuqori tartibli differentsiallari.

1⁰. Yuqori tartibli xususiy hosilalar.

Faraz qilaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funktsiya ochiq $E \subset R^m$ to'plamning har bir $x = (x_1, x_2, \dots, x_m) \in E$ nuqtasida

$$\frac{\partial f(x)}{\partial x_i} = f'_{x_i} \quad (i = 1, 2, \dots, m)$$

xususiy hosilalarga ega bo'lsin. Bu xususiy hosilalar $x_1, x_2, \dots, x_m$ o'zgaruvchilarning funktsiyasi bo'lib, ular ham xususiy hosilalarga ega bo'lishi mumkin:

$$\frac{\partial}{\partial x_k} \left(\frac{\partial f(x)}{\partial x_i} \right) = (f'_{x_i}(x))_{x_k} \quad (i, k = 1, 2, \dots, m)$$

Bu xususiy hosilalar berilgan $f(x)$ funktsiyaning ikkinchi tartibli xususiy hosilalari deyiladi va

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_i} \quad \text{ëku} \quad f''_{x_i x_k}(x) \quad (i, k = 1, 2, \dots, m)$$

kabi belgilanadi:

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_i} = f''_{x_i x_k}(x) = \frac{\partial}{\partial x_k} \left(\frac{\partial f(x)}{\partial x_i} \right).$$

Agar $i \neq k$ bo'lsa,

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_i}$$

ikkinchi tartibli xususiy hosila aralash hosila deyiladi.

Agar $i = k$ bo'lsa, ikkinchi tartibli xususiy hosilalar

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_i} = f''_{x_i x_k}(x)$$

quyidagicha

$$\frac{\partial^2 f(x)}{\partial x_i^2} = f''_{x_i^2}(x)$$

yoziladi.

$f(x)$ funksiyaning uchinchi, to'rtinchi va h.k. tartibdagi xususiy hosilalari xuddi yuqoridagiga o'xshash ta'riflanadi. Umuman, $f(x) = f(x_1, x_2, \dots, x_m)$ funksiyaning $x_1, x_2, \dots, x_{i_{n-1}}, x_{i_n}$ o'zgaruvchilari bo'yicha n -tartibli xususiy hosilasi berilgan funksiyaning $(n - 1)$ – tartibli xususiy hosilasi

$$\frac{\partial^{n-1} f(x)}{\partial x_{i_{n-1}} \partial x_{i_{n-2}} \dots \partial x_{i_1}} \quad (i_1 + i_2 + \dots + i_{n-1} = n - 1)$$

ning x_{i_n} o'zgaruvchi bo'yicha xususiy hosilasi sifatida ta'riflanadi:

$$\frac{\partial^n f(x)}{\partial x_{i_n} \partial x_{i_{n-1}} \dots \partial x_{i_2} \partial x_{i_1}} = \frac{\partial}{\partial x_{i_n}} \left(\frac{\partial^{n-1} f(x)}{\partial x_{i_{n-1}} \dots \partial x_{i_2} \partial x_{i_1}} \right).$$

Bu holda ham $i_1, i_2, \dots, i_n$ lar bir-biriga teng bo'lmaganda

$$\frac{\partial^n f}{\partial x_{i_n} \dots \partial x_{i_2} \partial x_{i_1}}$$

aralash hosila deyiladi.

Agar $i_1 = i_2 = \dots = i_n = k$ bo'lsa, n – tartibli xususiy hosilalar quyidagicha

$$\frac{\partial^n f(x)}{\partial x_k^n}$$

yoziyadi. Ushbu

$$\frac{\partial^2 f}{\partial x_k \partial x_i}, \quad \frac{\partial^2 f}{\partial x_i \partial x_k} \quad (i \neq k)$$

aralash hosilalar funksiyaning turli o'zgaruvchilari bo'yicha differensiallash tartibi bilan farq qiladi:

$$\frac{\partial^2 f}{\partial x_k \partial x_i}$$

da $f(x_1, x_2, \dots, x_m)$ funksiyaning avval x_i o'zgaruvchisi bo'yicha, so'ng x_k o'zgaruvchisi bo'yicha xususiy hosilasi hisoblangan bo'lsa,

$$\frac{\partial^2 f}{\partial x_i \partial x_k}$$

da esa avval x_k o'zgaruvchisi bo'yicha, so'ng x_i o'zgaruvchisi bo'yicha xususiy hosilasi hisoblangan. Ular bir-biriga teng ham bo'lishi mumkin, teng bo'lmasdan qolishi ham mumkin (misollar keyingi punktda keltiriladi).

Aralash hosilalarning tengligini quyidagi teorema ifodalaydi.

1-teorema. Faraz qilaylik, $f(x_1, x_2, \dots, x_m)$ funksiya $x^0 = (x_1^0, x_2^0, \dots, x_m^0) \in E \subset R^m$

nuqtada n marta differensiallanuvchi bo'lsin. U holda x^0 nuqtada $f(x_1, x_2, \dots, x_m)$ funksiyaning ixtiyoriy n -tartibli aralash hosilalarning qiymati $x_1, x_2, \dots, x_m$ o'zgaruvchilar bo'yicha qanday tartibda differensiallanishiga bog'liq bo'lmaydi.

◀ Bu teoremaning isboti, keyingi punktda ikki o'zgaruvchili funksiya uchun keltiriladigan teorema isboti kabi bo'ladi. ▶

2⁰. Yuqori tartibli differensialar.

Faraz qilaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funksiya ochiq $E \subset R^m$ to'plamda berilgan, $x \in E$ nuqtada ikki marta differensiallanuvchi bo'lsin.

1-ta’rif. $f(x)$ funksiya differensiali $df(x)$ ning differensial berilgan funksiyaning x nuqtadagi ikkinchi tartibli differensial deyiladi va $d^2 f(x)$ kabi belgilanadi:

$$d^2 f(x) = d(d f(x))$$

Endi funksiya ikkinchi tartibli differensialini uning xususiy hosilalari orqali ifodalanishini ko'rsatamiz.

Ravshanki,

$$d f(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_1} d x_1 + \frac{\partial f(\mathbf{x})}{\partial x_2} d x_2 + \dots + \frac{\partial f(\mathbf{x})}{\partial x_m} d x_m$$

bo'lib, $u^x = (x_1, x_2, \dots, x_m)$ ga hamda $dx_1, dx_2, \dots, dx_m$ larga bog'liq bo'ladi. Bu tenglikda $dx_1, dx_2, \dots, dx_m$ larni tayinlangan deb hisoblab, $df(x)$ ni $x_1, x_2, \dots, x_m$ o'zgaruvchilarning funksiyasi sifatida qarab, uning differensialini topamiz:

$$\begin{aligned}
d(f) &= d \left(\frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_m} dx_m \right) = \\
&= dx_1 \left[\frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) dx_1 + \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) dx_2 + \dots + \frac{\partial}{\partial x_m} \left(\frac{\partial f}{\partial x_1} \right) dx_m \right] + \\
&+ dx_2 \left[\frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) dx_1 + \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) dx_2 + \dots + \frac{\partial}{\partial x_m} \left(\frac{\partial f}{\partial x_2} \right) dx_m \right] + \\
&+ \dots + \\
&+ dx_m \left[\frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_m} \right) dx_1 + \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_m} \right) dx_2 + \dots + \frac{\partial}{\partial x_m} \left(\frac{\partial f}{\partial x_m} \right) dx_m \right] = \\
&= \frac{\partial^2 f}{\partial x_1^2} dx_1^2 + \frac{\partial^2 f}{\partial x_2^2} dx_2^2 + \dots + \frac{\partial^2 f}{\partial x_m^2} dx_m^2 + 2 \left[\frac{\partial^2 f}{\partial x_1 \partial x_2} dx_1 dx_2 + \right. \\
&\left. + \frac{\partial^2 f}{\partial x_1 \partial x_3} dx_1 dx_3 + \dots + \frac{\partial^2 f}{\partial x_1 \partial x_m} dx_1 dx_m + \frac{\partial^2 f}{\partial x_2 \partial x_3} dx_2 dx_3 + \frac{\partial^2 f}{\partial x_2 \partial x_4} dx_2 dx_4 + \dots \right]
\end{aligned}$$

$$+ \dots + \frac{\partial^2 f}{\partial x_2 \partial x_m} dx_2 dx_m + \dots + \frac{\partial^2 f}{\partial x_{m-1} \partial x_m} dx_{m-1} dx_m \Big] =$$

$$= \left(\frac{\partial}{\partial x_1} dx_1 + \frac{\partial}{\partial x_2} dx_2 + \dots + \frac{\partial}{\partial x_m} dx_m \right)^2 f.$$

Bunda simvollik ravishda yozilishidan foydalaniladi. U quyidagicha tushuniladi: qavs ichidagi yig'indi kvadratga ko'tarilib, so'ng f ga "ko'paytiriladi". Keyin daraja ko'rsatkichlari xususiy hosilalar tartibi deb hisoblanadi. Demak,

$$d^2 f(x) = \left(\frac{\partial}{\partial x_1} dx_1 + \frac{\partial}{\partial x_2} dx_2 + \dots + \frac{\partial}{\partial x_m} dx_m \right)^2 f. \quad (1)$$

$f(x)$ funksiyaning x nuqtadagi uchinchi, to'rtinchi va h.k. tartibdagi differensiallari ham yuqoridagidek ta'riflanadi.

Umuman, $f(x)$ funksiyaning x nuqtadagi $(n-1)$ -tartibli differensial $d^{n-1} f(x)$ ning differensial $f(x)$ ning n -tartibli differensial deyiladi va $d^n f(x)$ kabi belgilanadi:

$$d^n f(x) = d(d^{n-1} f(x)).$$

Agar, $f(x)$ funksiya x nuqtada n marta differensialanuvchi bo'lsa, u holda

$$d^n f(x) = \left(\frac{\partial}{\partial x_1} dx_1 + \frac{\partial}{\partial x_2} dx_2 + \dots + \frac{\partial}{\partial x_m} dx_m \right)^n f \quad (2)$$

bo'ladi.

3^o. Murakkab funksiyaning yuqori tartibli differensiallari.

Biz yuqorida funksiyaning yuqori tartibli differensiallarini bayon etdik. Unda funksiya argumenti $x_1, x_2, \dots, x_m$ lar erkli o'zgaruvchilar edi.

Aytaylik, $f(x) = f(x_1, x_2, \dots, x_m)$ funksiyada $x_1, x_2, \dots, x_m$ o'zgaruvchilarning har biri $t_1, t_2, \dots, t_k$ o'zgaruvchilarning funksiyalari bo'lsin ($x_i = \varphi_i(t_1, t_2, \dots, t_k)$).

Qaralayotgan $f(x)$ va $x_i = \varphi_i(t)$ ($i = 1, 2, \dots, m$) funksiyalar n marta differensiallanuvchilik shartlarini bajargan deb, murakkab $f(x(t))$ funksiyaning yuqori tartibli differensiallarini hisoblaymiz.

Ma'lumki, differensial shaklning invariantligi xossasiga binoan, murakkab funksiyaning differensial

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_m} dx_m$$

bo'ladi. Differensiallash qoidalaridan foydalanib funksiyaning ikkinchi tartibli differensialini topamiz:

$$\begin{aligned} d^2 f &= d(df) = d\left(\frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_m} dx_m\right) = \\ &= d\left(\frac{\partial f}{\partial x_1} dx_1\right) + d\left(\frac{\partial f}{\partial x_2} dx_2\right) + \dots + d\left(\frac{\partial f}{\partial x_m} dx_m\right) = \\ &= d\left(\frac{\partial f}{\partial x_1}\right) \cdot dx_1 + \frac{\partial f}{\partial x_1} d(dx_1) + d\left(\frac{\partial f}{\partial x_2}\right) \cdot dx_2 + \frac{\partial f}{\partial x_2} d(dx_2) + \\ &+ \dots + d\left(\frac{\partial f}{\partial x_m}\right) \cdot dx_m + \frac{\partial f}{\partial x_m} d(dx_m) = \\ &= d\left(\frac{\partial f}{\partial x_1}\right) \cdot dx_1 + d\left(\frac{\partial f}{\partial x_2}\right) dx_2 + \dots + d\left(\frac{\partial f}{\partial x_m}\right) dx_m + \\ &+ \frac{\partial f}{\partial x_1} d^2 x_1 + \frac{\partial f}{\partial x_2} d^2 x_2 + \dots + \frac{\partial f}{\partial x_m} d^2 x_m = \\ &= \left(\frac{\partial}{\partial x_1} dx_1 + \frac{\partial}{\partial x_2} dx_2 + \dots + \frac{\partial}{\partial x_m} dx_m\right)^2 f + \\ &+ \frac{\partial f}{\partial x_1} d^2 x_1 + \frac{\partial f}{\partial x_2} d^2 x_2 + \dots + \frac{\partial f}{\partial x_m} d^2 x_m. \end{aligned}$$

Demak,

$$d^2 f = \left(\frac{\partial}{\partial x_1} dx_1 + \frac{\partial}{\partial x_2} dx_2 + \dots + \frac{\partial}{\partial x_m} dx_m \right)^2 f + \\ + \frac{\partial f}{\partial x_1} d^2 x_1 + \frac{\partial f}{\partial x_2} d^2 x_2 + \dots + \frac{\partial f}{\partial x_m} d^2 x_m . \quad (3)$$

Shu yo'l bilan berilgan murakkab funksiyaning keyingi tartibdagi differensiallari topiladi.

1-eslatma. (1) va (3) formulalarni solishtirib, ikkinchi tartibli differensiallarda differensial shaklning invariantligi xossasi o'rinli emasligi ko'ramiz.

2-eslatma. Agar $f(x_1, x_2, \dots, x_m)$ funksiya argumentlari $x_1, x_2, \dots, x_m$ larning har biri $t_1, t_2, \dots, t_k$ o'zgaruvchilarning chiziqli funksiyasi bo'lsa, u holda $f(x)$ funksiyaning ikkinchi tartibli (umuman, n -tartibli) differensial differensial shaklning invariantlik xossasiga ega bo'ladi.

◀ Aytaylik,

$$x_i = b_i + a_{i_1} t_1 + a_{i_2} t_2 + \dots + a_{i_k} t_k \quad (i = 1, 2, \dots, k)$$

bo'lsin. U holda, ravshanki,

$$d^2 t_1 = d^2 t_2 = \dots = d^2 t_k = 0$$

bo'lib,

$$d^2 x_i = a_{i_1} d^2 t_1 + a_{i_2} d^2 t_2 + \dots + a_{i_k} d^2 t_k = 0$$

bo'ladi. (3) formuladan foydalanib topamiz:

$$d^2 f = \left(\frac{\partial}{\partial x_1} dx_1 + \frac{\partial}{\partial x_2} dx_2 + \dots + \frac{\partial}{\partial x_m} dx_m \right)^2 f .$$

Bu esa $d^2 f$ ning (1) formula ko'rinishiga ega ekanligini bildiradi. ▶

Mashqlar

1. Ushbu

$$u = f(x, y), \quad x = t^2 + s^2, \quad y = t \cdot s$$

funksiyaning ikkinchi tartibli differensialini topilsin.

2. Ushbu

$$u = y \cdot \varphi(x^2 - y)$$

funksiya quyidagi

$$\frac{1}{x} \frac{\partial u}{\partial x} + \frac{1}{y} \frac{\partial u}{\partial y} = \frac{u}{y^2}$$

tenglikni qanoatlantirishi isbotlansin.

Adabiyotlar

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